



USN

--	--	--	--	--	--	--	--	--	--

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of NITTE (Deemed to be University)
First Semester M.Tech. (CBCS) Degree Examinations
January 2023

22CCT131 - MECHANIZATION IN CONSTRUCTION

Duration: 3 Hours

Max. Marks:100

Note: Answer Five full questions choosing Two full questions from Unit – I & Unit – II each and One full question from Unit – III.

Unit – I		Marks	BT*	CO*	PO*
1. a)	An equipment has a purchase value of Rs 30, 00,000 and a useful life of 5 years and a salvage value of 3, 00,000. Determine total depreciation, book value for all 5 years by i) Constant percentage Method. ii) Straight line method.	10	L*4	1	1,2,4
b)	Compare the following with examples: i) Standard equipments and special equipments. ii) Write a note on factors affecting the performance of equipment in site.	10	L3	2	1,5
2. a)	Determine the probable cost of owning and operating cost of construction equipment with following data: Engine capacity=90 Hp, Crankcase capacity = 25 liters. Time between oil change = 90 hours. Cost of equipments = 20 lakhs. Useful life = 5 years. Hours usage per years=3000. Salvage value = 2, 00,000. Operating factor = 0.6 Maintenance cost = 80% of depreciation cost. Cost of diesel =Rs 80/ liter. Cost of Lubricating oil =Rs 175/ liter.	10	L4	2	1,2,4
b)	Write a note on power shovel with the help of neat sketch.	10	L1	1	1
3. a)	Write a short note on following i). Classification of Rippers ii). Ripping efficiency	10	L1	1	1
b)	List the factors affecting the output of dragline.	10	L1	2	1,5
Unit – II					
4. a)	Explain the steps involved in aggregate production.	10	L2	3	1,6,7
b)	Explain with the help of a neat sketch the various components of bridge with their functions.	10	L2	4	1,6,7
5. a)	Write a note on equipments used for cutting in rebar fabrication.	10	L1	3	1,6,7,8
b)	Explain Box pushing technology for tunneling.	10	L2	4	1,6,7
6. a)	Briefly Explain i). Segregation of concrete mix ii). Concrete pumps in concrete production.	10	L2	3	1,6
b)	Explain Pipe Jacking method in Trenchless Technology with neat sketches.	10	L2	4	1,6,7,8
Unit – III					
7. a)	Explain safety and environmental issues involved in mechanization in construction.	10	L2	5	1,6,7,8
b)	Briefly Explain i) Prefabricated construction. ii) Different types of scaffolding systems.	10	L2	5	1,6,8
8. a)	Explain any two types of wall form work based on the operations.	10	L2	5	1,6,8
b)	Write a note on types of accidents in construction industry.	10	L1	5	1,6,7,8

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
